

Nahid Hasan Sagar

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PROFESSIONAL SUMMARY

M.S. candidate in Statistical Science at Indiana University Bloomington with experience in reproducible research, quantitative methods, and high-volume data workflows. Skilled in R, Python, SQL, Git, and Bash, with a strong interest in evidence synthesis, meta-analysis, and AI-assisted research pipelines for development economics and policy evaluation.

EDUCATION

Indiana University, College of Arts and Sciences

Bloomington, IN

Master of Science in Statistical Science

Expected May 2027

Thesis: Time-Series Analysis Using Bayesian Additive Regression Trees (BART)

Relevant Coursework: Introduction to Statistical Computing; Applied Statistical Computing; Fundamentals of Statistics I–II; Applied Linear Models I–II; Statistical Consulting; Bayesian Modeling; Research in Statistics

University of Dhaka

Dhaka, Bangladesh

Bachelor of Science in Physics

August 2024

EXPERIENCE

Research Assistant, StatMIND Lab

2025 – Present

Indiana University Bloomington

Bloomington, IN

- Developed efficient C++ implementations using Rcpp and RcppArmadillo for high-dimensional neuroimaging workflows, improving computational performance and enabling faster analysis of large-scale scientific data.
- Reproduced original R implementation results and validated correctness through unit testing, benchmarking, and quality checks to ensure robust and reproducible outputs.
- Built benchmarking workflows to compare runtime, scalability, and computational efficiency across implementations, achieving 4x–5x speedups in core analytic computations.
- Worked with structured data, reproducible analysis pipelines, and documentation practices that support reliable research workflows and cross-team collaboration.

Teaching Assistant, Department of Statistics

2025 – Present

Indiana University Bloomington

Bloomington, IN

- Support undergraduate statistics courses through instruction, discussion sections, and individualized guidance on statistical reasoning and computation.
- Grade approximately 200 assignments per cycle and provide detailed feedback that emphasizes interpretation, model assumptions, and correct methodological practice.
- Conduct office hours and exam-review sessions that strengthen students' conceptual understanding and technical problem-solving skills.

SELECTED RESEARCH & PROJECTS

M.S. Thesis: Time-Series Analysis Using Bayesian Additive Regression Trees

R, Bayesian Modeling, Reproducible Research

- Conducted thesis research on Bayesian machine learning methods for time-series analysis under the supervision of Dr. Matthew T. Pratola.
- Applied statistical modeling, uncertainty quantification, and validation procedures to compare predictive performance on temporal data using reproducible workflows.

Ordinary Least Squares from Scratch

Python, NumPy, Pandas, Matplotlib

- Developed a transparent implementation of ordinary least squares from first principles, emphasizing matrix-based estimation and interpretation of model results.
- Validated outputs through numerical checks and visual diagnostics to ensure statistical correctness and clear communication of findings.

TECHNICAL SKILLS

Programming and Data Tools: R, Python, SQL, C++, C, MATLAB, Bash, Git, GitHub, LaTeX, Quarto, R Markdown, VS Code

Statistical Methods: Bayesian inference, BART, MCMC, time-series analysis, linear and logistic regression, mixed-effects models, experimental design, model validation, bootstrap methods, meta-analytic thinking

Research and Workflow Skills: Reproducible analysis, data cleaning and validation, benchmarking, literature review, technical writing, collaborative research documentation, quality assurance for quantitative projects

Languages: Bengali (native), English (fluent), Hindi (conversational), Urdu (conversational)

LEADERSHIP & HONORS

- Elected Graduate Vice Chairperson of the Statistics Club at Indiana University Bloomington for the 2026–2027 academic year.
- Represented Indiana University at the 2026 Velocity Conference at UC Berkeley as part of the science community.
- Collaborated in the development of a comprehensive online Linear Algebra curriculum for undergraduate learners during the COVID-19.